

2 Abaca plants are found in the Philippines.

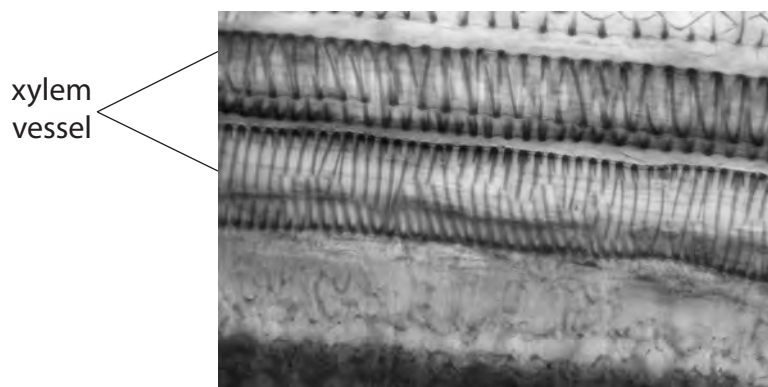
The photograph shows a mat made from abaca plant fibres.



(Source: © Mehmet Çetin / Alamy Stock Photo)

Plant fibres include xylem vessels.

(a) The photograph shows some xylem vessels, as seen using a light microscope.



(Source: © STEVE GSCHMEISSNER / SCIENCE PHOTO LIBRARY)

A stain was added to the xylem before viewing using a microscope.

Give a reason why it is usual to stain specimens in microscopy.

(1)

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- (b) Explain how the arrangement of cellulose molecules and secondary thickening in xylem vessels contributes to the physical properties of the cell wall.

(4)

(Total for Question 2 = 5 marks)

